

THE SELF-DRIVING PROJECT

FOUR AUTONOMOUS LABORATORIES

FROM OPERATIONS RESEARCH TO AUTONOMOUS FINANCIAL INSTITUTIONS

SELF-DRIVING MODELS: FROM CONTROL TO FINANCIAL JUDGMENT

Four Laboratories for Perception, Judgment, Action and Feedback

Race car, air traffic control, dynamic logistics and cybersecurity control room as a unified research program for transferring governed adaptive control to trading, multi-market portfolio management, capital allocation and systemic-risk management.

CENTRAL PROPOSITION

Classical control systems are powerful because they measure, optimize and act under explicit rules. Their weakness appears when the environment changes in ways that require interpretation: objectives conflict, priorities move, uncertainty changes meaning, and propagation creates consequences that are not local. Large language models can add a bounded layer of contextual judgment. The resulting architecture is not “LLM control.” It is a governed closed loop in which algorithms preserve measurement, feasibility and hard constraints while judgment makes the system more sensitive to circumstances.

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October 2026

Pedagogical Research Monograph

AI for Financial Practitioners / Autonomous Systems / Institutional
Design

The four computational examples are synthetic pedagogical laboratories. Their purpose is architectural transfer: to understand how governed judgment can complement deterministic control in financial institutions.

Abstract

This monograph develops a conceptual research program from four synthetic computational laboratories: a race-car simulator, a multi-airport air-traffic-control system, a dynamic delivery network, and a cybersecurity control room. The cases motivate a shared architecture in which a system perceives a changing environment, constructs an operational state, makes a bounded policy choice, authorizes and executes action, observes outcomes, and feeds the new state back into the next decision cycle. Deterministic constraints, validation, auditability and escalation surround that loop. The laboratories are pedagogical simulations, not evidence that language models improve financial performance or are ready for unsupervised deployment.

The common intellectual problem is the boundary between optimization and judgment. A classical algorithm can calculate a target speed, detect a predicted conflict, optimize a delivery route or score a cyber incident. Those calculations remain indispensable. Yet they assume that the meaning of the state, the ordering of objectives, and the relevance of contextual circumstances are sufficiently specified in advance. The four experiments deliberately stress that assumption. Rain changes what aggressive driving means. Weather and congestion transform the topology of airspace. An urgent medical shipment changes the priority structure of a delivery network. A cyber compromise propagates through dependencies so that a locally efficient response can become systemically dangerous. In these cases, intelligence requires more than recomputing the same objective function.

The proposed answer is *optimization with judgment*. A language model is treated as a bounded supervisory component, not a sovereign controller: it receives structured evidence, may recommend among pre-authorized alternatives, and remains subject to independent governance. The finance mapping is a research proposition, not a demonstrated result: race-car adaptation suggests regime-sensitive execution; air-traffic coordination suggests cross-portfolio and balance-sheet management; logistics suggests capital, liquidity and collateral triage; and cyber propagation suggests systemic-risk monitoring. The monograph’s contribution is to connect this architecture to portfolio choice, market microstructure, financial intermediation and model governance, then specify testable hypotheses and a business-validation agenda. It reports no empirical financial results.

THE UNIFYING RESEARCH QUESTION

How can a system preserve the reliability, speed and auditability of deterministic control while adding enough contextual judgment to recognize when changing circumstances require a different interpretation, priority or action?

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1. The Adventure: Four Laboratories, One Architecture

The four exercises were initially motivated by different operational problems. The race car asked how an autonomous machine can react at high frequency without confusing strategic intelligence with low-level control. Air traffic control raised the problem from one moving object to a network of interacting objects whose safe trajectories depend on weather, capacity, separation and sequencing. The delivery network introduced heterogeneous commitments and made priority explicit: the issue was no longer only where to send a vehicle, but which obligation should be served first when the environment changes. The cybersecurity control room introduced a fourth dimension—propagation—where the state of one node can alter the risk of many others and where local actions must be evaluated against system-wide consequences. Taken together, the examples form a deliberate progression. They move from **adaptation**, to **coordination**, to **reprioritization**, to **contagion-aware intervention**. This progression is more important than the surface domain. Each case exposes a different reason why a purely static or hard-coded policy can become insufficient even when its mathematics remains correct.

THE FOUR QUESTIONS

Race car: How should behavior change quickly when the operating regime changes?

Air traffic: How should many interdependent trajectories be coordinated under shared constraints?

Logistics: How should scarce resources be re-sequenced when priorities change?

Cyber control room: How should intervention change when risk propagates through a network?

The common denominator is not that an LLM “runs” the system. In all four experiments the most reliable architecture is layered. Deterministic functions remain responsible for geometry, arithmetic, physical limits, feasibility, authorization and execution. The language model is most valuable where the system must interpret a situation, reconcile competing objectives, or decide whether an unusual state should change the policy regime. Judgment is therefore an additional form of intelligence embedded inside a control architecture.

This distinction matters especially in finance. Financial institutions already possess sophisticated optimization engines. They do not need an LLM to replace a pricing model, a risk engine, an order-management system, a collateral optimizer or a settlement platform. The more interesting opportunity is to connect those systems through a judgment layer capable of interpreting circumstances while remaining bounded by mandates, limits and governance.

Why these four experiments belong to one research program

The laboratory sequence is best understood as a controlled comparison rather than a collection of illustrations. Each environment changes the meaning of an intelligent intervention while retaining a recognizable decision cycle. In the race car, the critical resource is response time. In air traffic control, it is shared capacity and the geometry of interactions. In logistics, it is the ordering of obligations under scarcity. In cybersecurity, it is the ability to interrupt a harmful process before its effects spread. These are distinct operational settings, yet all require the controller to distinguish the state of the world from the objectives governing its response.

The common scientific question is whether a judgment layer improves decisions relative to an appropriately

specified deterministic baseline. This comparison must be fair. The baseline should be permitted to use conventional optimization, predictive models, thresholds, event-driven rules and robust control. Otherwise an apparent gain from language-model reasoning could simply reflect a weak benchmark. The relevant claim is narrower and more defensible: when circumstances are difficult to encode completely in advance, a supervised model may help choose among admissible policies, interpret exceptions and identify when the prevailing priority hierarchy should be reconsidered.

This distinction suggests a testable hypothesis. Let the institution possess a set of feasible policies and an evolving operating context. A classical controller applies an established selection rule. A judgment-augmented controller is allowed to propose a different ordering of those same policies, subject to identical non-negotiable constraints. The question is whether the second arrangement improves realized outcomes after accounting for latency, error, escalation costs and governance overhead. The result cannot be inferred from the persuasiveness of an explanation; it must be evaluated from the sequence of states and actions actually produced.

The progression from local adaptation to systemic consequences

The first laboratory deliberately starts with a relatively legible physical environment. The car is one controlled object, and the effects of steering or braking become visible almost immediately. Its main lesson concerns time-scale separation: fast numerical control should not wait for slow deliberation. A supervisory model can adjust the driving posture while deterministic controllers preserve the immediate physical response. The corresponding financial insight is that interpretation of a market regime and placement of an order need not occur at the same frequency.

The second laboratory introduces interdependence. Aircraft do not simply follow independent optimal routes. A maneuver may alter separation, congestion and the feasible trajectories of other aircraft. Consequently, the state must describe the network and not merely its constituent objects. This is analogous to a financial institution in which one desk's hedge consumes market liquidity, one portfolio's financing demand affects the treasury function, and a decision to reduce exposure can create correlated execution pressures elsewhere. The coordinator's objective is to protect the integrity of the whole system rather than maximize the score of every component separately.

The third laboratory makes explicit an issue that is often concealed in optimization models: the objective function is itself an institutional choice. A delivery route can be mathematically optimal for travel time yet undesirable when a high-priority obligation becomes urgent. Replanning therefore involves more than updating distances. It can require a reasoned change in the ordering of commitments. In financial organizations this resembles liquidity triage, the allocation of scarce balance-sheet capacity and the sequencing of obligations under stress. A model that faithfully optimizes yesterday's priorities can be precisely wrong for today's institution.

The fourth laboratory adds propagation. A security event may begin locally but spread through credentials, services, dependencies or communication channels. The risk of a node depends partly on the network it can affect. In finance, this mechanism resembles counterparty exposure, funding contagion, payment-system dependencies and common-asset liquidation. The relevant intervention may have costs for an individual component while reducing the expected loss of the system. Thus the progression ends with a distinctly systemic criterion: evaluate an action by the distribution of consequences it may generate throughout the network.

The experimental discipline: baselines, authority and evidence

A serious interpretation of these exercises requires distinguishing three objects: the simulated environment, the decision policy and the institutional rules governing action. The environment produces observations and consequences. The policy recommends what to do. Governance decides whether the recommendation may be implemented. If those objects are merged, it becomes impossible to determine whether a favorable outcome came from better reasoning, a more permissive safety rule or an artificial simplification of the simulation.

The four notebooks therefore motivate a common experimental protocol. Begin with a reproducible scenario and a deterministic baseline. Introduce a bounded judgment component at an explicit point in the control loop. Require its outputs to satisfy a schema, pass independent checks and fall back safely when unavailable or invalid. Record the observations presented to the model, its proposed action, the governance decision, the executed action and the subsequent state. Finally, compare the two architectures across repeated disturbances rather than a single favorable run. This protocol makes the contribution of judgment observable and its limitations contestable.

The outcome measures should reflect the laboratory's particular objective. For the car, they include safety, progress, stability and responsiveness under changing grip or sensor confidence. For air traffic, they include conflict avoidance, throughput, delay and network resilience. For logistics, they include service completion, missed commitments, utilization and the treatment of exceptional priorities. For cybersecurity, they include propagation prevented, containment costs, false interventions and recovery time. A financial transfer should preserve this plural evaluation rather than collapse all outcomes into a convenient single performance number.

There is also an important epistemic limit. Synthetic simulations demonstrate architectural possibilities; they do not establish that an LLM is safe for unsupervised trading, aviation or security operations. Their value is methodological. They allow researchers to locate the decision boundary at which interpretation enters, make the associated authority explicit and design measurements of the incremental value of judgment. The resulting proposition is institutional as much as technological: intelligence can be distributed among specialized components, while accountability remains attached to defined rights, limits and observable actions.

Scope, contribution and limits

This document is a conceptual synthesis and research design, not a completed empirical finance paper. It introduces no proprietary market data, validated trading results, causal estimates or production-system evidence. Its central contribution is a falsifiable architecture: a contextual supervisor may select among pre-authorized policy regimes while quantitative engines retain measurement, optimization and execution, and governance retains authority. Claims of benefit are hypotheses to test against strong deterministic and human baselines. The phrase “self-driving financial institution” denotes a governed decision process, not a legal person, a fully autonomous bank or permission to transfer fiduciary, regulatory or board accountability to a model.

The financial objective is institution-specific. An asset manager must remain within client mandates and applicable duties; a bank treasury must manage funding, liquidity, capital and payment obligations; a broker or trading desk must account for execution quality, market impact, inventory and conduct rules. Systemic stability is a public-policy concern that cannot be reduced to one firm's private objective. Accordingly, no single scalar “financial objective” or universal action policy is assumed in this monograph.

2. The Canonical Architecture: Perception, Judgment, Action, Feedback

At the highest level, all four systems can be compressed into three principal layers plus feedback:



The simplicity is deceptive. Each box can contain several specialized components. Perception can include sensing, data validation, state estimation, forecasting and anomaly detection. Judgment can include deterministic scoring, policy selection, LLM reasoning, optimization and escalation. Action can include validation, governance, execution and actuator logic. Feedback can include telemetry, P&L, realized risk, service outcomes, incident progression and audit evidence.

Perception: converting observations into a decision state

A system cannot judge directly from the world. It requires a representation. The race car converts sensor measurements into estimates of position, lateral error, curvature, grip and confidence. Air traffic control converts aircraft telemetry, airport capacity and weather into predicted trajectories and conflict states. Logistics converts orders, deadlines, capacity, congestion and disruptions into slack, feasibility and service risk. Cybersecurity converts alerts, vulnerabilities, identity events, dependencies and anomalies into a risk state and a propagation map.

The important lesson is that perception is not merely data collection. It is the disciplined construction of the state on which action will depend. In finance the equivalent is the difference between receiving prices and understanding the institution's state: positions, liquidity, margin, concentrations, factor exposures, settlement obligations, counterparty dependencies and confidence in the underlying data.

Judgment: deciding what the state means for policy

Judgment begins where measurement ceases to be sufficient. It asks whether the current state changes what should matter. A wet track can make caution dominate lap time. A weather cell can make throughput subordinate to separation. An urgent medical parcel can outrank an economically efficient sequence. A cyber compromise at a highly connected node can justify containment that would appear excessive if the node were considered in isolation.

This is the natural location for bounded language-model reasoning. The model does not need to calculate what deterministic systems calculate better. Its comparative advantage is contextual synthesis: interpreting a structured situation, applying a hierarchy of objectives, explaining trade-offs, and recommending among already-valid alternatives. The word *bounded* is essential. The judgment layer should see enough to reason but should not automatically possess the authority to execute.

Action: converting a recommendation into an authorized intervention

Action is not synonymous with judgment. Between recommendation and execution sits governance. A race-car safety governor can reduce a proposed speed factor. An air-traffic harness can reject a maneuver that violates separation or runway rules. A delivery validator can refuse reassignment to a failed or overloaded vehicle. A cyber control room can require stronger authorization for isolation of critical infrastructure.

The financial analogue is familiar. A portfolio manager may recommend; risk limits constrain; compliance verifies; execution systems enforce market and account rules; settlement systems determine what actually completes. Autonomous finance should preserve this institutional separation rather than erase it.

Feedback: why autonomy is a trajectory, not a single answer

Once an action is executed, the world changes. The car's new position becomes the next observation. Aircraft trajectories shift. Delivery ETAs and capacity change. Cyber risk can fall, migrate or reveal new dependencies. The system must therefore operate as a loop, not as a sequence of isolated prompts.

This is a decisive conceptual difference between a chatbot and a self-driving institution. A chatbot can answer a question and stop. A control system changes the state it will subsequently observe. It must remember what it did, measure what happened, and decide again.

A minimal formal representation

The architecture can be expressed without turning the argument into a mathematical-control treatise. Let x_t denote the underlying state of the system at time t , y_t the observations available to the institution, and $\hat{x}_t$ the operational state constructed from those observations. Perception is therefore a mapping

$$\hat{x}_t = P(y_{0:t}),$$

where the history notation emphasizes that an intelligent state estimate may depend on more than the most recent observation. Let $\mathcal{A}(\hat{x}_t)$ be the set of actions that remain feasible under physical, legal, operational and risk constraints. Judgment does not receive an unconstrained universe of possibilities; it operates over this admissible set and contextual information c_t :

$$a_t^* = J(\hat{x}_t, c_t, \mathcal{A}(\hat{x}_t)).$$

A separate governance operator G then determines whether the recommendation can become an authorized action u_t :

$$u_t = G(a_t^*, \hat{x}_t, \mathcal{R}),$$

where $\mathcal{R}$ represents hard rules, delegated authority, limits and escalation requirements. Finally, the system evolves according to $x_{t+1} = F(x_t, u_t, w_{t+1})$, with w_{t+1} collecting disturbances that were not under the controller's authority. The next observation closes the loop.

This notation clarifies the institutional proposition. The LLM need not estimate every state, solve every optimization problem or execute every action. Its distinctive contribution can be concentrated in J : interpreting context, reconciling objectives and selecting among already admissible responses. The reliability of the overall system depends equally on P , G , F and the quality of feedback. Judgment is therefore a component of autonomous control, not a synonym for autonomy itself.

THE CENTRAL DESIGN PRINCIPLE

LLM judgment should enrich the interpretation of a controlled system, not dissolve its control architecture. The more consequential the action, the more important it becomes to separate **intelligence** from **authority**.

Decision rights and the separation of time scales

A useful architecture distinguishes the frequency of observation, the frequency of policy revision and the frequency of execution. A race car can require steering corrections at intervals far shorter than those appropriate for a language-model consultation. Similarly, a trading engine may process market data and enforce limits continuously while a supervisory judgment component reviews regime changes only at scheduled or event-triggered moments. There is no requirement that all components operate at the same clock speed.

Let a fast controller operate at each time step while the supervisory policy parameter θ_k changes only at selected review times k . The executed action can be represented schematically as

$$u_t = G(C(\hat{x}_t; \theta_{k(t)}), \hat{x}_t, \mathcal{R}),$$

where C is the conventional controller and $k(t)$ identifies the latest authorized supervisory policy. The judgment layer can recommend a change to θ , but cannot bypass G . This formulation clarifies why a system can become more context-sensitive without introducing an unpredictable language-model response into every latency-sensitive transaction.

Decision rights must be specified as carefully as the mathematical mappings. A recommendation is not an authorization, an authorization is not an execution, and an execution is not a verified outcome. Each transition can fail or require escalation. In a financial institution, this separation is familiar from investment mandates, risk limits, compliance approval, trade execution and post-trade reconciliation. Autonomous architecture should make those distinctions executable, not merely describe them in a governance policy.

Uncertainty, objectives and the value of judgment

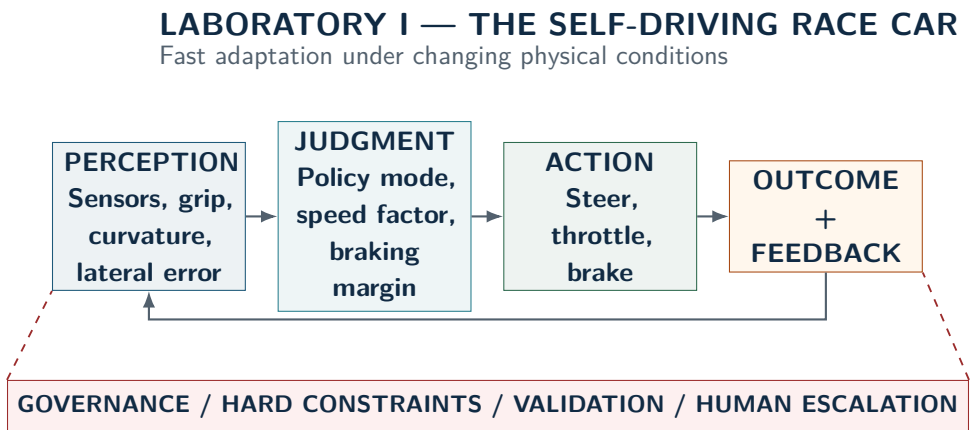
The operational state is rarely known perfectly. Observations may be delayed, incomplete, noisy or inconsistent. An intelligent system should therefore retain a measure of confidence alongside its estimate. Low confidence can change the admissible action set, trigger conservative defaults or require a human decision. A confident but mistaken interpretation is especially dangerous when it silently relaxes an institutional constraint.

Judgment becomes valuable when the controller faces multiple legitimate objectives whose relative importance depends on circumstances. Safety, timeliness, cost, continuity, fairness and systemic stability need not share one permanent set of weights. However, contextual flexibility must not become discretionary rule-breaking. A practical design distinguishes hard constraints from negotiable priorities. The former define the feasible region; the latter help select a policy within it. This distinction permits adaptation while preserving the rule of law, contractual commitments and operational safety.

Finally, feedback should measure not only whether the immediate objective was achieved, but also whether the system behaved as intended. Audit records should connect state estimates, policy recommendations, approvals, rejected proposals, executed actions and realized outcomes. Such records support model-risk management, retrospective evaluation and institutional learning. They also make it possible to ask the central empirical question: did contextual judgment add value beyond an excellent deterministic system, and was that value sufficient to justify its costs and risks?

3. Laboratory I — The Race Car: Adaptation at High Frequency

The race-car laboratory provides the most transparent entry point into the research program because the causal chain from observation to consequence can be seen almost physically. A vehicle occupies a position on a track, travels at a measurable speed, faces curvature and grip constraints, receives imperfect sensor information, and must continuously transform that information into steering, throttle and braking. Yet the pedagogical objective of the laboratory is not vehicle engineering. It is to expose the architecture of a self-driving system in a setting where the distinction between calculation, judgment, authority and execution cannot easily be hidden behind institutional language.



Operational problem and architecture

The notebook deliberately decomposes the apparent act of “driving” into specialized layers. Sensors produce noisy observations of position, heading, speed and track conditions. A perception and estimation layer transforms those observations into an operational state: lateral error, curvature, estimated grip, sensor confidence and other quantities that can be used by the controller. A deterministic planning layer computes physically meaningful targets. Fast lateral and longitudinal controllers then generate numerical steering, throttle and brake proposals. These components remain inspectable and mechanically constrained.

The language model is introduced at a different time scale. It does not calculate steering angles at every simulation tick. Instead, it receives a compact description of the operating situation and recommends a bounded strategic posture: a policy mode, speed factor, braking margin or overtaking stance. Validation normalizes and bounds that recommendation. A deterministic safety governor remains between recommendation and execution, with authority to clip, reshape or veto commands when curvature, grip, uncertainty or emergency conditions make the proposed posture unsafe. Telemetry records both the proposed and executed actions, making the chain of causation auditable.

This separation is analytically important. A system may be highly capable at interpretation without being granted direct authority over actuators. Conversely, a simple deterministic governor can possess

decisive practical authority because of its architectural position. The laboratory therefore distinguishes *intelligence* from *control rights*. The LLM may influence policy, but it does not own the plant, the safety envelope or the final execution gate.

A compact formalism: fast control inside a slower policy regime

Let x_t denote the vehicle state and $\hat{x}_t$ its estimate. A conventional controller computes a fast action

$$u_t = C(\hat{x}_t, r_t; \theta_t),$$

where r_t is the physical reference trajectory and θ_t collects policy parameters such as desired aggressiveness, speed scaling and braking margin. The key contribution of judgment is not to replace C but to update θ_t at a slower supervisory cadence:

$$\theta_{k+1} = J(\hat{x}_k, z_k, q_k), \quad k \in \{0, m, 2m, \dots\},$$

where z_k represents contextual conditions such as rain or an unusual event and q_k represents confidence in perception. A safety operator S then produces the executable command $\tilde{u}_t = S(u_t, \hat{x}_t, \mathcal{K})$, with $\mathcal{K}$ denoting hard physical and safety constraints. The formal distinction is modest but powerful: deterministic control answers *how* to move under a policy; judgment helps determine *which policy* is appropriate under current circumstances.

What the laboratory teaches about judgment

The dry-to-rain transition makes the distinction observable. The equations of vehicle motion do not cease to apply when grip falls. What changes is the meaning of an acceptable operating posture. A speed that was reasonable under dry conditions can become aggressive under rain. The same lateral error can demand a different response when the margin for recovery has narrowed. Sensor degradation adds a second dimension: the system can become uncertain about the state precisely when the state is becoming more dangerous. A mature controller must therefore react not only to estimated risk but also to confidence in the estimate.

This is where hard-coded rules alone encounter an important design problem. One can always add thresholds: if grip is below a specified value, reduce speed by a fixed percentage; if confidence is below another threshold, increase braking margin. Such rules are useful and should remain available. But as contextual dimensions multiply, the policy table becomes increasingly brittle. Judgment becomes valuable when several weak signals, exceptional conditions and competing objectives must be interpreted together. The LLM can synthesize those conditions and recommend a posture while the deterministic stack continues to enforce the invariants that should never be negotiable.

The laboratory therefore suggests a hierarchy of intelligence. Measurement should be precise where precision is available. State estimation should be quantitative. Fast control should remain deterministic where latency and stability matter. Judgment should enter where the question is contextual and policy-like. Governance should remain independently executable. This hierarchy avoids two symmetrical errors: treating every decision as a numerical optimization problem, or treating every numerical control problem as a task for generative intelligence.

Transfer to finance: algorithmic trading as regime-sensitive control

The financial analogue is algorithmic trading, particularly systems that must react rapidly while remaining inside explicit risk and execution constraints. Market data, order books, volatility, liquidity, inventory and exposure correspond to the race car's sensor stream. State estimation becomes the construction of a market and portfolio state. The fast controller becomes the pricing, quoting, hedging or execution algorithm. Position limits, order-size constraints, compliance rules and kill switches become the safety governor. The judgment layer operates above these mechanisms and asks whether the current market regime justifies a different policy envelope.

The analogy is especially useful because it discourages the superficial proposition that an LLM should simply “trade.” A market-making engine can calculate quotes faster and more reliably than a conversational model. An execution algorithm can enforce participation constraints mechanically. A risk engine can calculate exposures from authoritative data. The more plausible role for an LLM is supervisory: interpreting whether a volatility shock, liquidity withdrawal, venue outage, correlation break, news event or unusual combination of signals warrants a more defensive or more selective posture.

Formally, a trading engine may continue to solve

$$u_t^{\text{trade}} = C(\hat{x}_t^M; \theta_t),$$

where $\hat{x}_t^M$ contains market and inventory state. Judgment updates θ_t rather than directly manufacturing arbitrary orders. It might recommend lower inventory tolerance, wider spreads, reduced participation, a different hedging posture or temporary suspension. Those recommendations remain subject to mandates and hard limits. The resulting architecture preserves latency-sensitive automation while adding contextual sensitivity at the level where context is economically meaningful.

The transfer also clarifies feedback. Trading actions alter inventory, liquidity consumption, market impact and P&L; in sufficiently large strategies they can alter the market state itself. The controller must therefore observe realized fills and consequences rather than treat a recommendation as the end of the process. The system is judged over a trajectory of decisions, not by whether a single generated explanation sounded plausible.

FINANCE TRANSFER I

Primary analogue: algorithmic and high-frequency trading.

Transferred capability: rapid regime adaptation without surrendering deterministic execution and risk control.

Role of judgment: change the policy envelope when liquidity, volatility, confidence or institutional context changes the meaning of the same numerical signals.

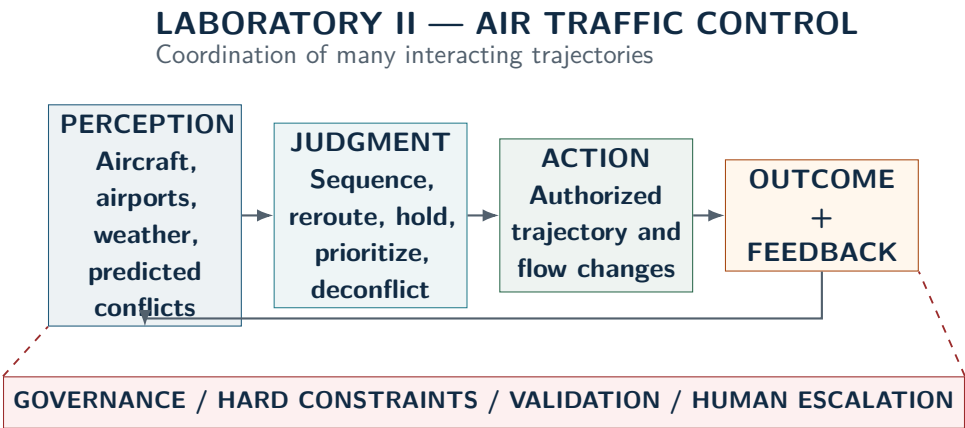
Governance principle: the judgment layer may recommend posture; limits, compliance and execution systems retain authority.

The race-car laboratory is therefore the foundational case. It demonstrates that adding judgment does not require replacing classical control. On the contrary, the most credible autonomous architecture becomes more layered as intelligence increases. The fast loop remains hard, quantitative and constrained; the slower loop becomes more interpretive; and governance determines how far interpretation is permitted to

influence action. In finance, this is the difference between putting an LLM in the order path and using an LLM to make an already sophisticated trading stack more sensitive to regime, uncertainty and context.

4. Laboratory II — Air Traffic Control: Coordination Across a Complex System

The air-traffic-control laboratory raises the problem from adaptive control of one object to coordinated control of many interacting objects. Aircraft have individual trajectories, destinations and constraints, but they occupy a shared system: airspace, airports, runways, weather corridors and sequencing capacity. The action taken for one aircraft can change the feasible choices available to others. This makes the laboratory particularly valuable for finance because large financial institutions are not collections of independent optimizers. They are networks of portfolios and businesses that compete for common capital, liquidity, collateral, market depth and operational infrastructure.



Operational problem: the controlled object is the system

In the simulator, perception begins with aircraft states, airport configurations, weather and network capacity. Prediction extends the current picture forward: where are aircraft likely to be, which trajectories may conflict, which airports may become congested, and how will weather alter the feasible network? Candidate actions include holding, rerouting, resequencing, changing flows and escalating exceptional situations. Deterministic rules preserve separation and other operational invariants. Supervisory judgment interprets congestion, weather and competing objectives without being allowed to repeal the safety architecture.

The decisive conceptual step is that a locally attractive action can be globally inferior. A direct route may save time for one aircraft while consuming scarce capacity needed by several others. Delaying one departure may reduce congestion downstream. Rerouting one flight can create a new conflict elsewhere. The state variable of interest is therefore not simply a vector of independent aircraft positions. It includes relationships, predicted intersections and shared bottlenecks. The controller must reason about the configuration of the network.

This introduces an institutional form of externality. Each trajectory imposes constraints on other trajectories. The problem is not solved by making every local controller smarter in isolation. Intelligence

must also exist at the coordinating layer. The ATC case therefore moves the research program from *adaptive autonomy* to *coordinated autonomy*.

A compact formalism: interacting trajectories and common capacity

Let x_t^i be the state of aircraft i , and let $X_t = (x_t^1, \dots, x_t^N)$ denote the system state. If each aircraft independently chose an action a_t^i to minimize its own cost $c_i(a_t^i, x_t^i)$, the resulting vector A_t could violate common constraints. The feasible set is joint:

$$A_t = (a_t^1, \dots, a_t^N) \in \mathcal{F}(X_t, W_t, K_t),$$

where W_t captures weather and K_t shared capacity. A system controller can be represented as choosing among feasible joint responses according to

$$A_t^* = J(X_t, \hat{X}_{t:t+h}, W_t, K_t, \mathcal{F}_t),$$

subject to lexicographic safety constraints that are not traded against throughput. The forecast horizon h matters because conflicts are often visible before they become immediate. Judgment is therefore partly anticipatory: it interprets a future configuration, not merely the current snapshot.

The formulation also highlights multiple time scales. Separation assurance may run continuously. Flow management may update at a slower interval. Strategic recovery from a major weather disruption may operate more slowly still. There is no architectural reason to force all three functions into the same model or cadence. A governed autonomous system is naturally asynchronous.

Judgment under shared constraints

The LLM's contribution is strongest when several legitimate objectives remain after hard safety constraints have been enforced. Throughput, delay, fairness, airport utilization and recovery speed may conflict. A purely scalar objective can encode these priorities, but doing so requires designers to specify their relative weights in advance for every relevant contingency. In unusual states, the question becomes institutional: which objective should dominate now, and why?

Bounded judgment can interpret the situation and select among already valid flow-management alternatives. Importantly, the model should not invent an unsafe trajectory simply because it produces a persuasive narrative. Candidate generation, conflict checking and authorization remain deterministic. This preserves a separation between *reasoning about the system* and *permission to alter the system*.

The ATC laboratory also makes auditability more demanding. In a single-car system one can inspect the causal chain for one object. In a coordinated network, the rationale for an action may depend on avoided conflicts or preserved capacity elsewhere. An audit trail should therefore record not only what was done but which system-level constraint or predicted interaction motivated the decision. Rejected alternatives become informative because they reveal the opportunity cost of coordination.

Transfer to finance: multi-market and multi-portfolio institutional control

The financial analogue is a bank, asset manager or investment platform coordinating many portfolios, strategies, venues and balance-sheet uses simultaneously. Positions resemble aircraft in the limited sense that each has its own trajectory and local economics while sharing scarce infrastructure. Capital, liquidity, collateral, risk appetite, funding capacity and settlement channels correspond to constrained airspace and runway capacity. Volatility, market closures, venue outages and funding stress correspond to weather.

Predicted conflicts become forecasts of concentration, liquidity gaps, margin calls, crowding, settlement bottlenecks and correlated drawdown.

The analogy reveals why optimizing each portfolio separately can produce a poor institutional outcome. Two desks can both be individually within limits and still rely on the same funding source. Several strategies can be independently attractive while crowded into the same factor or liquidity pool. Hedging one portfolio can increase basis risk elsewhere. Posting collateral to protect one obligation can reduce the institution's capacity to meet another. Local optimality does not imply system optimality.

A coordinating judgment layer therefore asks questions above the level of the individual optimizer. Which market or desk should receive scarce risk capacity? Which exposures should be reduced first when liquidity deteriorates? Should a profitable strategy be slowed because several other businesses depend on the same funding channel? Does a venue outage require rerouting orders, changing hedges, conserving liquidity, or reducing activity altogether? The answer can depend on mandate, urgency, substitutability and the expected reaction of other participants.

A useful financial representation is

$$A_t^F \in \mathcal{F}(P_t, L_t, C_t, R_t, O_t),$$

where P_t denotes positions, L_t liquidity, C_t collateral and capital resources, R_t risk limits and O_t operational constraints. Specialized engines can generate candidate actions for each portfolio. The supervisory layer evaluates the joint configuration and recommends a coordinated response. Governance then verifies that every action remains inside legal mandates, risk appetite and delegated authority.

This is more than diversification. Diversification describes properties of exposures; coordinated control describes the dynamics of actions. In stressed markets, selling an asset changes market impact, posting collateral changes liquidity, drawing a credit line changes funding capacity, and closing a hedge changes another exposure. The institution participates in the system it is trying to control. Feedback is endogenous.

FINANCE TRANSFER II

Primary analogue: multi-market, multi-asset, multi-portfolio and balance-sheet control.

Transferred capability: coordination of interacting trajectories under shared constraints.

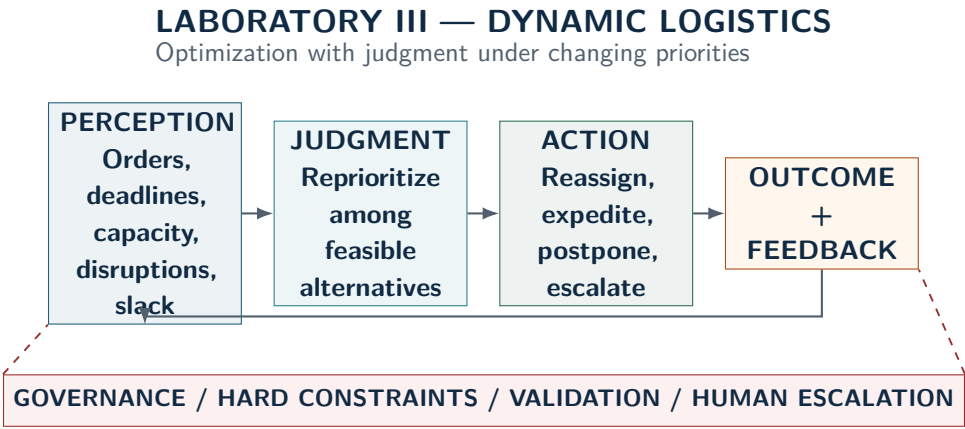
Role of judgment: choose system-level priorities when locally attractive actions compete for common capacity.

Governance principle: safety and hard institutional constraints are lexicographic; judgment operates only inside the admissible system-wide action set.

The ATC laboratory therefore changes the unit of analysis. The relevant object is no longer the strategy or portfolio but the institution-wide configuration. This is an important step toward self-driving finance. A genuinely autonomous financial institution would require not only intelligent desks but an intelligent coordinating layer capable of seeing common bottlenecks, anticipating interactions and modifying the aggregate policy while preserving hard constraints. The transfer from airspace to markets is precisely the transfer from local optimization to governed system coordination.

5. Laboratory III — Dynamic Logistics: Priorities, Triage and Re-Sequencing

The dynamic-logistics laboratory begins closest to classical operations research. A fleet, packages, routes, deadlines and capacity constraints define an optimization problem with a recognizable objective: allocate vehicles and sequence deliveries efficiently while satisfying service commitments. The research question appears only after the initial plan is produced. Congestion, weather, vehicle failures, urgent arrivals and customer changes alter the environment. Some disruptions merely require recalculation. Others change which commitments deserve priority. The distinction between these two cases is the intellectual center of the laboratory.



From re-optimization to reprioritization

A conventional optimizer is conditional on a specified state, objective function and feasible set. If travel times change, the system can re-optimize. If a vehicle fails, capacity can be updated and routes recomputed. These are important forms of adaptation, but they do not necessarily require judgment. The harder case occurs when the shock changes the institutional meaning of the commitments themselves. An urgent medical package, a protected service class, a near-expired deadline or a customer exception can make the original sequence inappropriate even if it remains technically feasible.

The notebook therefore separates measurement and feasibility from priority. Deterministic code estimates travel time, slack, capacity, delay risk and cost. Candidate generation constructs a bounded menu of feasible responses such as retain, reassign, expedite, postpone where permitted, or escalate. A deterministic baseline ranks those alternatives under a fixed policy. The LLM receives the same structured case and the finite candidate set, then applies an institutional hierarchy to recommend among them. A governance gate independently checks the recommendation before execution. Feedback reconstructs the network state after each action.

The baseline is essential for research design. Without it, one cannot determine whether the language model added judgment or merely reproduced a ranking already implied by the coded score. The laboratory

therefore makes the incremental contribution of judgment testable: compare the selected action with the deterministic counterfactual and examine the states in which the choices diverge.

A compact formalism: when the objective hierarchy itself becomes state dependent

Suppose the deterministic planner evaluates feasible action a using K criteria $C_k(a; s_t)$ and fixed weights w_k :

$$a_t^0 = \arg \min_{a \in \mathcal{A}(s_t)} \sum_{k=1}^K w_k C_k(a; s_t).$$

Ordinary re-optimization updates s_t while leaving the institutional hierarchy w unchanged. Judgment becomes relevant when a contingency z_t changes the appropriate priority structure:

$$a_t^J = \arg \min_{a \in \mathcal{A}(s_t)} \sum_{k=1}^K w_k(s_t, z_t) C_k(a; s_t),$$

possibly subject to lexicographic priorities that should not be reduced to compensating weights. The important variable is therefore not only the optimal action but the rule governing which objectives dominate. An LLM can help infer a context-sensitive priority vector or choose among policy regimes, while deterministic code preserves $\mathcal{A}(s_t)$ and validates the resulting action.

This formulation explains why the laboratory is about judgment rather than simply a more complicated optimization model. In principle, every contingency could be encoded ex ante in a sufficiently elaborate objective function. In practice, institutions operate with incomplete specifications, qualitative policies and exceptional states. The question is how to introduce flexibility without allowing discretion to become unbounded.

Triage as a normal institutional capability

Triage is often associated with emergencies, but the logistics experiment suggests a broader interpretation. Any institution with scarce resources and heterogeneous commitments is continuously performing some form of triage. The issue is whether the ordering is implicit and rigid or explicit, contextual and governed. A delivery network reveals the problem clearly because service classes and deadlines are visible. In organizations, the equivalent priorities may be dispersed across budgets, mandates, board decisions, contractual obligations, regulatory requirements and strategic plans.

Language models are useful precisely because they can read and synthesize those heterogeneous forms of policy. But the architecture must distinguish policy interpretation from resource authority. The model can explain that an urgent commitment should dominate an ordinary one; it should not be able to transfer funds or override legal restrictions merely because its rationale is persuasive. Candidate generation and governance define the boundary of discretion.

The laboratory also introduces a useful concept of *institutional regret*. A plan can remain feasible and even locally efficient while becoming undesirable relative to the institution's changed priorities. Measuring performance only against the original objective can therefore reward the wrong behavior. Evaluation should ask whether the system recognized the contingency, changed priorities when appropriate, preserved hard commitments, and returned toward normal policy once the contingency passed.

Transfer to finance: dynamic budgeting and scarce-resource allocation

The transfer to finance is unusually broad because financial institutions are full of allocation and sequencing problems. A corporate budget allocates capital across projects. A bank allocates liquidity and balance-sheet capacity across businesses. Treasury allocates collateral. A credit committee allocates risk capacity. A holding company allocates capital across subsidiaries. Senior management allocates attention and implementation capacity. In each case, an initial plan is optimized under assumptions that can later change.

Consider a capital budget with projects $i = 1, \dots, n$, each characterized by expected return, cash requirement, completion state, contractual commitment, strategic importance and risk. A static optimizer may produce an allocation b^0 . After a shock, simply feeding updated cash flows into the same objective may be insufficient. A cyber incident may make resilience investment temporarily dominant. A near-completion project may deserve protection because abandoning it destroys option value. A regulatory deadline may become binding. A liquidity shock may make cash preservation more important than expected return. The institution needs a mechanism for changing the ordering of objectives, not merely updating the numbers.

The same logic applies to bank liquidity. Under normal conditions, carry and efficient use of the balance sheet matter. Under stress, settlement certainty, collateral availability and survival can dominate. In credit, a portfolio may normally prioritize risk-adjusted return, while a restructuring episode may elevate recovery, legal sequencing or systemic-client continuity. In all these cases, the central question is: *when should a feasible plan cease to govern because the institution's priorities have changed?*

A self-driving allocation system can therefore be organized in layers. Quantitative engines maintain authoritative measures of cash, exposure, capital, collateral and contractual constraints. Optimization engines construct feasible alternatives. Judgment interprets the contingency and recommends a priority regime. Governance checks delegated authority and protected constraints. Execution occurs in the systems of record. Feedback observes realized liquidity, completion, risk and service outcomes.

FINANCE TRANSFER III

Primary analogue: budgeting, capital allocation, liquidity, collateral and credit triage.

Transferred capability: governed reprioritization of scarce resources after contingencies.

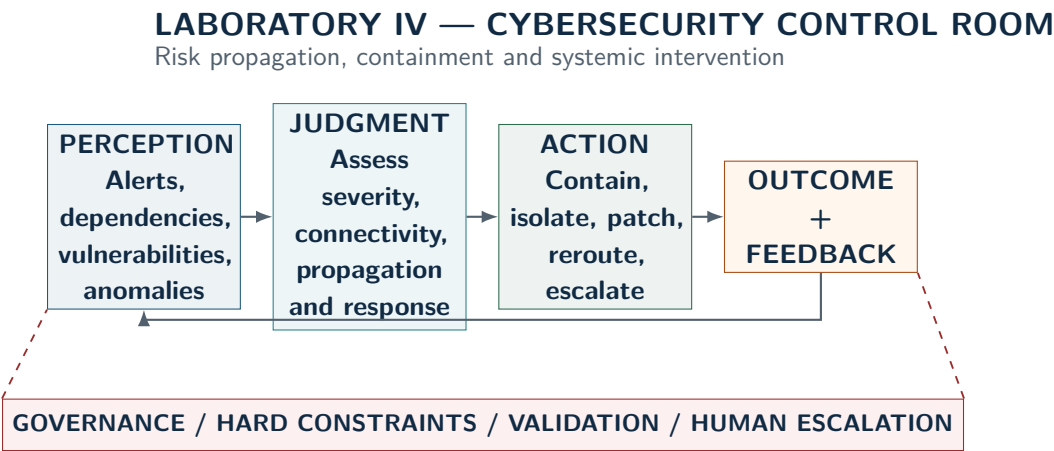
Role of judgment: recognize when a plan that remains feasible is no longer the plan the institution should want.

Governance principle: context may alter priorities, but it does not erase legal constraints, mandates, protected commitments or delegated authority.

The logistics laboratory is therefore where the research program most clearly crosses from control engineering into institutional design. A race car changes posture; an air-traffic system coordinates trajectories; a logistics institution must decide what matters most. This is the closest analogue to executive decision-making. The central contribution of LLM judgment is not better arithmetic but the capacity to interpret a changed environment against a hierarchy of purposes. The central contribution of governance is to make that interpretive flexibility safe, bounded and auditable.

6. Laboratory IV — Cybersecurity Control Room: Propagation and Systemic Risk

The cybersecurity control-room laboratory adds a feature that is qualitatively different from the previous cases: propagation. In the race car, a disturbance changes the condition of one controlled object. In air traffic, actions interact through shared capacity. In logistics, contingencies alter priorities across scarce resources. In cyber risk, an adverse state can travel through a dependency network. A compromised identity, service, vendor or network segment can change the probability and consequence of compromise elsewhere. The relevant object of control is therefore a dynamic graph of exposures rather than a collection of isolated alerts.



Operational problem: risk acquires meaning through connectivity

A classical security stack can classify alerts, score severity and execute predefined playbooks. These capabilities remain indispensable. The control-room experiment asks what happens when the same local signal has radically different implications depending on where it occurs. An anomaly on a peripheral device may be containable with limited intervention. A similar anomaly involving privileged identity, shared authentication, payment infrastructure or a critical service can create a very different system state because of the paths available for propagation.

Perception must therefore combine event evidence with topology. The operational state includes not only which nodes appear stressed but how nodes depend on one another, which services are critical, what privileges connect them, and where uncertainty remains high. Judgment evaluates severity jointly with connectivity and possible future spread. Candidate actions can include isolation, credential rotation, service degradation, rerouting, patching, containment or escalation. Hard governance is especially important because defensive actions can themselves create damage. Isolating a critical service may stop propagation while interrupting legitimate activity.

The laboratory thus contains two propagation processes: the threat can propagate, and the consequences of the defense can propagate. A locally aggressive response may reduce cyber risk while increasing

operational risk. Judgment must reason over this trade-off without being granted unilateral authority over critical infrastructure.

A compact formalism: propagation over a dependency graph

Let the system be represented by a graph $\mathcal{G}_t = (V, E_t)$ with node-risk vector $r_t \in [0, 1]^n$. Let W_t be a weighted dependency matrix whose element $w_{ij,t}$ represents the strength of a channel through which stress at node j can affect node i . A stylized propagation equation is

$$r_{t+1} = \sigma(\alpha r_t + \beta W_t r_t + e_{t+1} - B(a_t)),$$

where e_{t+1} represents new evidence or shocks, $B(a_t)$ the risk reduction associated with an intervention, and $\sigma(\cdot)$ bounds the state. The expression is intentionally schematic: its purpose is to make clear that tomorrow's risk depends on both local risk and network transmission.

A purely local policy chooses a_t from the current severity of a node. A propagation-aware policy evaluates an action against a system loss function such as

$$L(a_t) = \sum_i \omega_i \mathbb{E}[r_{i,t+h} \mid r_t, W_t, a_t] + C_{\text{op}}(a_t),$$

where ω_i reflects criticality and C_{op} captures the operational cost of containment. Judgment becomes valuable when the topology, evidence quality, criticality and institutional consequences must be interpreted together. Deterministic graph analytics and risk scores quantify the state; the judgment layer interprets what the pattern means for intervention.

Why propagation changes the control problem

The distinction between node risk and system risk is fundamental. A node can be individually small yet systemically important because it is highly connected or sits on a critical path. Conversely, a severe local event can be relatively contained if dependencies are limited. The control room must therefore distinguish magnitude from centrality, and current loss from potential transmission.

This creates a natural role for scenario reasoning. The system can ask not only “how severe is this alert?” but “what becomes reachable if this hypothesis is correct?” and “what legitimate operations would be impaired by the proposed containment?” Such questions are difficult to collapse into one universal threshold. They combine structured evidence with contextual interpretation. An LLM can synthesize the narrative of the incident and the dependency map, but authoritative telemetry, identity records and graph calculations should remain outside the model.

The governance lesson is equally important. Increasing intelligence should not imply increasing trust. A model capable of understanding a sophisticated threat can still be wrong, manipulated or uncertain. Critical controls should therefore remain independent: authorization, segregation of duties, immutable logging, reconciliation and escalation. The architecture is designed so that a failure of judgment does not automatically become a failure of institutional control.

Transfer to finance: contagion and systemic-risk management

The financial transfer is especially strong because systemic risk is fundamentally a propagation problem. Banks, funds, clearing members, payment infrastructures, funding markets and asset classes are connected through credit exposures, collateral chains, common holdings, liquidity dependence, settlement obligations

and confidence. An institution can appear sound in isolation while occupying a network position that amplifies stress. Likewise, a shock that is small in accounting terms can become consequential through feedback loops.

The cyber mapping is direct. A compromised node corresponds to a stressed institution, market or infrastructure. Dependency edges correspond to interbank claims, collateral relationships, common asset holdings, payment links or shared funding sources. Lateral movement corresponds to contagion through withdrawals, margin calls, fire sales, replacement hedges or information effects. Containment corresponds to exposure reduction, liquidity provision, ring-fencing, margin adjustment, trading restrictions, resolution tools or other authorized interventions. Telemetry becomes supervisory reporting, payment flows, market data and high-frequency liquidity indicators.

A stylized financial propagation model can be written similarly:

$$s_{t+1} = \Phi(s_t, \Lambda_t, p_t, a_t, \varepsilon_{t+1}),$$

where s_t is the vector of institutional stress, Λ_t the network of exposures and dependencies, p_t market prices, and a_t authorized interventions. The key point is not the particular functional form. It is that intervention should be evaluated against the future system state, including endogenous responses. Selling assets to improve one balance sheet can depress prices and weaken others. Raising margin can protect a clearing mechanism while accelerating liquidity demand. Providing liquidity can arrest contagion but alter incentives. The system is reflexive.

An LLM judgment layer can add value by combining network analytics with qualitative information: news, operational events, legal restrictions, supervisory priorities and scenario narratives. But it should not invent balance-sheet data, calculate authoritative capital ratios from prose, or possess unilateral intervention authority. Quantitative systems measure; graph models expose channels; scenario engines estimate transmission; judgment interprets; governance authorizes.

FINANCE TRANSFER IV

Primary analogue: systemic-risk surveillance, contagion management, operational resilience and crisis control rooms.

Transferred capability: propagation-aware intervention in a network of dependencies.

Role of judgment: recognize when local stress has systemic meaning and compare interventions by their network consequences.

Governance principle: the more powerful the intervention, the stronger the independent authorization, evidence and audit requirements.

The cybersecurity laboratory completes the progression of the four cases. It moves the self-driving framework from reacting to the current state toward reasoning about how states propagate through a connected system. This is precisely the conceptual bridge required for systemic finance. The autonomous institution must not only perceive its own condition and choose its own action; it must anticipate how stress and intervention can move across counterparties, markets and infrastructures. Judgment becomes valuable because propagation changes the meaning of local evidence. Governance becomes indispensable because the remedies themselves can have systemic consequences.

7. The Transfer Function: From Operations Research to Finance

The four laboratories can be summarized as four transfer functions. They are not metaphors in the loose sense; they are architectural correspondences. The same categories of information, judgment, authority and feedback recur even though the physical meaning changes.

Laboratory	Operational capability	Financial transfer
Race car	Fast adaptation to regime and uncertainty	Algorithmic trading; dynamic risk posture; regime-sensitive execution
Air traffic control	Coordination of interacting trajectories under shared constraints	Multi-market, multi-asset, multi-portfolio and balance-sheet coordination
Dynamic logistics	Re-sequencing scarce resources under changing priorities	Budgeting, capital allocation, liquidity, collateral and credit triage
Cyber control room	Propagation-aware containment in a dependency network	Systemic risk, contagion, operational resilience and crisis intervention

The transfer becomes even clearer when expressed by layer:

Layer	Operations/control	Finance	Governance question
Perception	Sensors, telemetry, alerts, disruptions	Prices, positions, exposures, liquidity, news, obligations	Is the state reliable and sufficiently complete?
Judgment	Policy mode, sequencing, prioritization, containment	Regime interpretation, capital priorities, systemic assessment	What should matter now?
Action	Steering, rerouting, dispatch, isolation	Orders, hedges, transfers, limits, interventions	Is the action authorized and feasible?
Feedback	Motion, traffic flow, service outcome, incident state	P&L, fills, liquidity, realized risk, contagion indicators	Did the intervention improve the system?

The deeper transfer is from a static optimization mindset to a closed-loop institutional mindset. Traditional finance often separates model construction from decision governance. The self-driving perspective forces them into one architecture. A forecast has value only if it enters a decision process; a decision has meaning only if it is authorized; an action has consequences that must be observed; and those consequences become the next state.

Financial foundations and scope conditions

The finance transfer rests on established problems in portfolio choice and market design, not on analogy alone. Mean–variance portfolio selection formalizes the trade-off between expected return and risk [16]; continuous-time portfolio theory makes the opportunity set and state evolve over time [17]. In execution, transaction costs and market impact mean that an action changes both the portfolio and the price at which it can be implemented [14]. Funding liquidity and market liquidity can reinforce one another during stress, making an institution’s response part of the risk mechanism it is trying to manage [15]. These foundations explain why a financial control problem must represent objectives, frictions, constraints and feedback explicitly.

Let s_t include positions, cash and funding, collateral, liquidity, exposures, market conditions and the institution’s mandate. Let a_t represent trades, hedges, collateral transfers or resource allocations. A

stylized dynamic allocation problem is

$$V_t(s_t) = \max_{a_t \in \mathcal{A}(s_t)} \mathbb{E}_t [r_{t+1}^{\text{gross}}(s_t, a_t) - \text{TC}_t(a_t) - \text{FC}_t(s_t, a_t) + \beta V_{t+1}(s_{t+1})],$$

with $s_{t+1} = F(s_t, a_t, \varepsilon_{t+1})$. Here TC denotes transaction and market-impact costs, while FC collects funding and balance-sheet costs. The admissible set $\mathcal{A}(s_t)$ must encode binding legal mandates, capital and liquidity limits, execution constraints and delegated authority; tail-risk constraints, such as a limit on expected shortfall, can be added where appropriate. This is a schematic decision problem, not a calibrated valuation model, and each institution must specify its own objective and constraints.

A bounded language-model supervisor should therefore select a pre-authorized policy regime m_t from a finite set $\mathcal{M}$, not invent the objective or directly emit unconstrained trades:

$$m_t = J(\hat{s}_t, c_t, \mathcal{P}) \in \mathcal{M}, \quad a_t = C_{m_t}(\hat{s}_t), \quad \tilde{a}_t = G(a_t, \hat{s}_t, \mathcal{R}).$$

A quantitative controller C_{m_t} computes the action, and an independent governance function G authorizes or rejects it. The economic value of this additional layer is conditional: it must improve decisions relative to robust optimization, state-dependent rules or expert review after accounting for costs and risk. Where those alternatives perform as well or better, an LLM adds no financial value.

The private and social objectives also differ. A firm's mandate and risk appetite do not internalize every spillover through common asset sales, shared funding, market impact or correlated model use. Systemic-risk monitoring therefore requires system-level measures and public oversight in addition to each firm's private optimizer. No private LLM can, by itself, resolve that governance problem.

8. What LLM Judgment Adds — and What It Must Not Replace

The phrase “LLMs add judgment” requires precision. A language model does not possess institutional judgment merely because it can produce fluent explanations. Judgment emerges only when the model is placed inside an architecture that supplies evidence, objectives, candidate actions, constraints, authority boundaries and feedback.

Rival explanations and failure modes

Contextual complexity alone is not a case for a language model. Robust control, Bayesian decision analysis, regime-switching models, rule-based exception handling and human escalation are credible alternatives and must be included as benchmarks. A language model may be confidently wrong, sensitive to prompt and retrieval choices, inconsistent across repeated runs, vulnerable to contaminated inputs, slower than operational deadlines, or dependent on a concentrated vendor. Fluent explanation is neither calibration nor causal evidence. The burden is to show incremental decision value, not merely that the model can summarize an event.

Where language models are genuinely useful

Their comparative advantage appears when several conditions coexist. The evidence is heterogeneous; not all objectives reduce naturally to one scalar; context changes the relevance of rules; exceptions matter; explanations are valuable; and a finite set of legitimate alternatives can be constructed. This is exactly the margin exposed by the four laboratories.

In the race car, judgment adjusts strategic posture rather than steering geometry. In ATC, it interprets network conditions rather than calculating every trajectory. In logistics, it reprioritizes among feasible actions rather than solving vehicle capacity. In cyber, it interprets propagation and response trade-offs rather than replacing topology or deterministic access control.

What should remain deterministic

Hard physical, legal, accounting and institutional constraints should remain outside the LLM wherever practical. Arithmetic should be calculated, not narrated. Feasibility should be checked by code. Permissions should be enforced by systems of record. Risk limits should be executable. Candidate identifiers should be validated. Model output should be treated as untrusted until it passes a governance gate.

This leads to a useful design equation:

$$\begin{aligned} \text{Autonomous intelligence} = & \text{Deterministic control} \\ & + \text{Bounded judgment} + \text{Independent governance} \\ & + \text{Feedback.} \end{aligned}$$

No term is sufficient by itself. Deterministic control without judgment can become brittle. Judgment without control can become unreliable. Governance without feedback can become ceremonial. Feedback without explicit state and objectives can become noise.

JUDGMENT IS NOT DISCRETION WITHOUT LIMITS

The objective is not to make a financial institution “more discretionary.” It is to make its adaptive reasoning explicit, inspectable and bounded. The LLM should expand sensitivity to circumstances while the architecture preserves limits, permissions, auditability and human authority.

9. A General Model of Governed Adaptive Control

Let s_t denote the perceived state at time t , constructed from observations o_t . Let $A(s_t)$ be the set of feasible actions produced by deterministic rules and constraints. A classical optimizer chooses

$$a_t^* = \arg \min_{a \in A(s_t)} C(a; s_t),$$

where C is a specified objective. This is powerful when the objective is stable and sufficiently complete.

The self-driving experiments suggest a richer structure. Suppose there are K legitimate objectives with weights w_k that can depend on state, uncertainty and contingency z_t :

$$a_t^* = \arg \min_{a \in A(s_t)} \sum_{k=1}^K w_k(s_t, z_t) C_k(a; s_t).$$

The difficult problem is often not computing a_t^* . It is determining whether the weights, ordering or lexicographic priority of objectives should change. Judgment operates at this meta-control level.

A bounded judgment function J can therefore be written as

$$\pi_t = J(s_t, z_t, \mathcal{P}, \mathcal{M}, A(s_t)),$$

where $\mathcal{P}$ represents policies and priorities and $\mathcal{M}$ represents relevant memory or institutional context. The output π_t is not necessarily an executable action. It can be a policy mode, ranking, recommendation or escalation decision. A governance function G then determines whether the proposed action is authorized:

$$\tilde{a}_t = G(a_t^*, \pi_t, s_t),$$

and execution changes the environment:

$$s_{t+1} = F(s_t, \tilde{a}_t, \varepsilon_{t+1}).$$

The loop repeats.

This formalization clarifies the division of labor. Optimization remains inside the architecture. Judgment can alter policy or priorities but does not abolish feasibility. Governance remains a separate mapping. The environment remains stochastic. Feedback makes the system adaptive.

10. From Autonomous Models to Autonomous Financial Institutions

The final transfer is institutional. A self-driving financial institution should not be imagined as one model connected to an execution API. It should be imagined as a federation of specialized systems operating at different time scales and with different authority.

At the perception layer, market data, accounting systems, risk engines, operational systems and external information construct a governed belief state. At the judgment layer, specialized models and LLM supervisors interpret regime, exceptions, priorities and interactions. At the action layer, optimizers construct feasible interventions while permissions, limits, compliance and human approvals determine what can be executed. At the feedback layer, transactions, P&L, exposures, service outcomes and incidents update the state and create an audit trail.

The four laboratories suggest four capabilities that a mature institution may need:

1. **Reflex mode** for fast, deterministic response—the race-car lesson.
2. **Coordination mode** for interacting portfolios and shared constraints—the ATC lesson.
3. **Triage mode** for scarce resources and changing priorities—the logistics lesson.
4. **Contagion mode** for propagation through networks—the cyber-control-room lesson.

These modes should not be implemented as four isolated applications. They are perspectives on the same institutional state. A liquidity event, for example, can require all four. Fast controls may reduce exposures; cross-portfolio coordination may prevent desks from competing for cash; triage may protect critical payments; contagion analysis may identify counterparties or markets through which stress can spread.

This is why the apparently simple three-layer architecture is so powerful. Perception, judgment and action are not a toy abstraction. They are a stable decomposition across domains. What changes is the state representation, the cadence, the objective hierarchy, the candidate actions and the governance boundary.

11. Design Principles for Finance

The experiments support a compact doctrine for building autonomous financial systems.

1. KEEP THE FAST LOOP FAST

Do not ask a language model to perform numerical or latency-sensitive control that deterministic systems already perform well. Use LLMs at supervisory cadences or exception points where contextual synthesis has value.

2. MAKE THE STATE EXPLICIT

Judgment should operate on a structured, auditable representation of the situation. Separate observations, estimates, forecasts and confidence. A model should not have to infer the institution's state from an uncontrolled mass of text.

3. CONSTRAIN THE ACTION SPACE

Where consequences are material, generate feasible candidate actions before the LLM reasons over them. This preserves operational validity and makes recommendations testable.

4. SEPARATE RECOMMENDATION FROM AUTHORITY

The component best able to explain a situation need not be the component allowed to execute. Validation, limits, permissions, segregation of duties and human escalation should remain independent.

5. PRESERVE A BASELINE

A deterministic policy is both a fallback and a counterfactual. Without it, the institution cannot measure whether the judgment layer actually improved decisions.

6. TEST THE LOOP, NOT THE ANSWER

Evaluate trajectories: stability, interventions, rejected actions, realized outcomes, recovery and propagation. A plausible one-shot recommendation is not evidence of autonomous competence.

Supervisory alignment and model scope

Financial governance must be precise about jurisdiction and date. For US banking organizations, the interagency guidance SR 26-2, issued 17 April 2026, supersedes SR 11-7. It is risk-based and expected to be most relevant to organizations with more than \$30 billion in assets but potentially relevant to smaller firms with significant model risk, and is expressly nonbinding and nonprescriptive. It excludes generative and agentic AI from its scope, but explicitly says that each banking organization's existing risk-management and governance practices should guide the determination of appropriate controls for systems outside the document [18]. The agencies have also indicated that a forthcoming request for information will address model risk, including AI. This scope boundary is neither a universal AI rule nor an exemption from other applicable obligations, and the guidance does not establish that a particular AI architecture is compliant. As of 7 October 2026, the FSB's June 2026 document on responsible AI adoption remains a consultation report rather than an adopted international standard [22]. These sources should inform, not replace, institution-specific legal and supervisory review.

12. Research Agenda: Measuring the Value of Judgment

The finance transfer is a set of hypotheses, not a result. The empirical question is whether bounded contextual judgment improves institutionally relevant outcomes relative to strong quantitative and human alternatives, after transaction, funding, operating and governance costs. The research should begin with a narrow use case and expand only when each additional layer of autonomy is justified by evidence.

Testable hypotheses

1. **Conditional value of judgment.** Relative to strong deterministic policies, a bounded LLM supervisor should add value primarily in pre-identified episodes involving policy ambiguity, interacting objectives or distribution shifts. In stable regimes, its incremental value may be zero or negative after latency and review costs.
2. **Governance trade-off.** Independent validation and authorization should reduce severe constraint violations and unauthorized actions, while potentially increasing false escalations, delay and missed opportunities. Report both sides of this trade-off rather than treating additional approvals as costless.
3. **System-level coordination.** Joint allocation across portfolios and balance-sheet functions may improve institution-wide liquidity and tail-risk outcomes even when it does not maximize each desk's standalone return. Test both local and aggregate effects, including spillovers caused by execution and common exposures.
4. **Net economic value.** The judgment layer is commercially and institutionally justified only if its incremental, risk-adjusted benefit exceeds model, data, integration, oversight, validation, resilience and incident costs. A fluent explanation or an improvement in a single headline metric is insufficient.

A staged empirical design

The first finance study should focus on a decision with explicit alternatives and measurable constraints, such as treasury liquidity and collateral triage during funding or market stress. This is a more defensible starting point than claiming that a language model generates alpha. Positions, cash, collateral, payment obligations, market conditions and policy documents should be represented with point-in-time provenance and confidence. Where institution data cannot be published, the study should disclose the synthetic data-generating process, assumptions and limits to external validity.

Compare at least four arms under identical information, action sets and hard constraints: (i) a fixed rule or current operating policy; (ii) a well-tuned state-dependent optimizer or regime-switching policy; (iii) an expert or existing human workflow where it can be consistently recorded; and (iv) a bounded LLM supervisor whose output selects only among pre-specified candidate actions. Include ablations that remove retrieval, governance or candidate restrictions in offline simulation only, to identify which component contributes to performance. Do not make an unconstrained agent a production comparator.

Use rolling, strictly out-of-sample evaluation with time-stamped data, frozen model and prompt versions, and explicit controls for look-ahead, survivorship, repeated testing and data leakage. Pair each policy against the same historical replay and simulated shocks. Stress cases should include abrupt volatility or funding changes, partial or contradictory observations, venue or counterparty outages, and networked fire-sale scenarios. Historical replay alone is not sufficient where actions change prices, liquidity or other institutions' states; combine it with a transparent market-impact or agent-based simulator, then state which causal channels the simulator cannot identify.

Outcomes, inference and falsification

Report net outcomes, not gross model recommendations. Depending on the use case, measures should include net return or operating cost, transaction and market-impact cost, funding and collateral cost, expected shortfall, drawdown, liquidity shortfall and duration, payment or settlement failures, limit exceptions, time to resolution, false intervention, escalation rate, fallback frequency and human-review time. No single Sharpe ratio, service score or average reward is an adequate objective for all institutional mandates.

Use paired scenario comparisons, confidence intervals and block-based resampling appropriate to serial dependence; pre-specify primary outcomes and subgroup analyses by regime; correct for multiple comparisons; and publish null and adverse results. Evaluate calibration and consistency separately from outcome quality. In shadow deployment, log the evidence, candidate set, model recommendation, authorization, action and realized consequence without allowing the recommendation to execute. Any later live experiment should begin with reversible, low-impact actions and an approved control group; high-impact interventions require a separate legal, risk and supervisory review.

The research program should be falsifiable. It should reject the business case if a tuned non-LLM policy matches the outcomes at lower cost, if apparent gains vanish after realistic impact and funding costs, if improvements depend on leaked information, or if error severity and common-mode failures exceed the institution's risk tolerance. This standard distinguishes a finance contribution from a technology demonstration.

13. Business Agenda: From Research to a Finance Offering

The commercial thesis is not “an LLM that trades.” It is a governed decision-orchestration layer that connects authoritative financial data, existing optimizers, institutional policy and human approval. This distinction matters economically and operationally. Cross-jurisdictional supervisory work describes current financial AI use as concentrated in internal operations and regulatory compliance, while revenue-generating applications remain limited [19]. The near-term business case should therefore emphasize measurable workflow, liquidity or control improvement, not promise excess returns.

Initial customer and product wedge

A credible first customer is a bank treasury, asset manager’s liquidity function or broker-dealer risk team with a recurring exception workflow and a named budget owner such as the treasurer, chief risk officer or operations executive. The initial use case could be liquidity and collateral triage: consolidate a governed state, surface policy-relevant events, rank a deterministic menu of feasible responses, provide evidence-linked rationale, and route the recommendation to an authorized decision-maker. It should begin in read-only or shadow mode, not by placing orders, moving client assets or altering regulatory limits.

The product would be an integration and control plane rather than a general-purpose chat interface. Its differentiating capabilities would be source-linked state construction; retrieval of versioned policies and mandates; candidate generation through approved quantitative systems; schema, permission and limit checks; human escalation; tamper-evident decision records; and replay tools that compare proposed, approved and executed actions against a baseline. The system should be model-agnostic so a financial institution can replace a foundation-model provider without rewriting its control framework.

Business model and unit economics

Three commercialization paths merit testing. An institution may first deploy the capability internally and capture verified reductions in process time or risk exposure. A vendor may license an enterprise control plane on a recurring basis, with separately scoped integration and validation services. A specialist provider may offer policy mapping, evaluation and ongoing assurance as a managed service. These are business hypotheses, not established market sizes or revenue forecasts; procurement cycles, data access, integration burden and supervisory scrutiny must be measured in pilots.

A disciplined value model is

$$\text{NPV}_\Delta = \sum_{t=0}^T \frac{\mathbb{E}_0[\Delta B_t - \Delta C_t]}{(1+r)^t},$$

where ΔB_t is the attributable operating, risk or capacity benefit relative to a documented counterfactual, and ΔC_t is the corresponding cost. The cost term includes model, data, integration, validation, oversight, resilience, capital and incident costs. The discount rate r is the institution’s appropriate hurdle rate. Benefits should be estimated against a documented counterfactual and avoid double-counting prevented losses, freed liquidity and staff time. Costs must include model inference, data rights, systems integration, independent validation, compliance review, human escalation, fallback operations, cyber protection and potential incident remediation. Outcome-based pricing tied to “avoided losses” should be treated cautiously because the counterfactual is difficult to observe and can create perverse incentives; predictable enterprise licensing plus transparent implementation fees is a more testable starting hypothesis.

Commercial stage gates and principal risks

A responsible go-to-market sequence is: (1) retrospective workflow and data audit; (2) reproducible offline benchmark against current and optimized baselines; (3) shadow operation with no authority to act; (4) advisory recommendations with explicit human approval; and (5) narrowly scoped execution only after independent validation, legal and compliance review, operational resilience testing and board-delegated approval. Each stage needs predefined continuation and stop criteria. Scale only if measured net value is positive, constraints remain intact, outage fallback works and the institution can reconstruct decisions end to end.

The business also faces risks that are themselves part of the product thesis. Financial Stability Board analysis identifies third-party concentration, data quality and governance, cyber risk and common-model behavior as potential channels through which AI can amplify vulnerabilities [20]. Its subsequent monitoring work highlights data gaps and supply-chain concentration as continuing surveillance challenges [21]. Vendor lock-in, confidential-data exposure, prompt or retrieval manipulation, model drift, correlated recommendations, liability allocation and uncertain supervisory treatment can erase projected savings. As of 7 October 2026, the FSB's responsible-AI document cited here is a consultation report, not a binding global standard; US banking guidance SR 26-2 also does not cover generative or agentic AI within its model definition. Institutions therefore need jurisdiction-specific review rather than marketing claims of regulatory approval.

The practical business agenda is consequently to sell verified control quality, integration and evidence—not autonomy as a slogan. A defensible offering earns expansion from low-risk decision support into more consequential workflows only as customer-specific evaluation demonstrates incremental value and governance remains effective. If those gates fail, the correct outcome is to retain the existing deterministic process.

14. Conclusion — Judgment as a Testable Layer Between Optimization and Autonomy

The four laboratories motivate a common control architecture: observe a changing state, distinguish measurement from priority, select only among feasible actions, assign authority independently, and use realized outcomes to update the next decision. The finance transfer is strongest where this loop already has quantitative state, hard constraints and meaningful interactions—execution, portfolio coordination, liquidity and collateral allocation, and propagation-aware risk management.

Finance also imposes requirements that a cross-domain analogy cannot supply by itself. Portfolio choice is intertemporal and mandate-dependent; trading incurs market impact and funding costs; actions alter the state being controlled; and private incentives do not internalize all systemic spillovers. These are not details to add after an LLM prototype. They define the objective, state, feasible set and evaluation standard from the outset.

This monograph has therefore advanced a research and business proposition, not established empirical superiority. A bounded contextual supervisor may help in specific regimes where policy interpretation matters, but it must compete against strong optimization, rule-based and human baselines and justify its total cost. The appropriate end state is not unrestricted autonomy. It is a measurable, institution-specific allocation of intelligence, execution and authority, with a deterministic fallback and a clear path to stop.

FINAL TAKEAWAY

The working proposition is **control with judgment**: preserve quantitative finance and independent governance, add contextual supervision only where rigorous out-of-sample evidence shows net value, and treat every expansion of authority as a separate business and risk decision.

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